



Unit 4 · Outcome 2 · Energy Sources

Where does the energy actually go?

KK05

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Energy is never used up — it's transformed, and degraded

The big idea Energy is never used up — it's transformed, and degraded A power station doesn't make energy. It converts energy that's already stored in something (chemical bonds in coal, water held high behind a dam, moving air) into the form we want: electrical energy . The First Law of Thermodynamics says energy can't be created or destroyed — only changed from one form to another. So if you put 100 units of energy in, 100 units come out. The catch is which forms they come out as. We want electricity. What we usually also get is a pile of low-grade heat

Efficiency = useful out ÷ total in

The one formula Efficiency = useful out ÷ total in This is the single equation behind every calculation in this dot point. Memorise it, and memorise that the answer can never exceed 100% . η = useful energy output / total energy input $\times 100\%$ η ("eta") = efficiency. Works with energy (J, kJ, MJ, GJ) or power (W, kW, MW) — as long as both top and bottom use the same units. Energy or power — same formula A question might give you energy (joules — a total amount) or power (watts — a rate, joules per second). The formula is identical. Just keep the units consis

The conversion-chain lab

Interactive · Scene 02 The conversion-chain lab Pick a generation source at Yarrabin and watch energy packets flow through the chain. Watch how many survive to the grid — and how many peel off as heat at every step. Energy flow Brown coal Gas (CCGT) Solar PV Wind Hydro Energy in 1000 MJ Useful electricity out — Wasted as heat — Conversion chain — Brown coal · overall efficiency — Energy input 1000 MJ Each box shows the energy surviving after that conversion. The overall efficiency is the running product — multiply the step efficiencies, never add them.

Multi-step: why the losses compound

The exam-critical skill Multi-step: why the losses compound When energy passes through several conversions, you multiply the step efficiencies (as decimals) — you never add or average them. $\eta_{\text{overall}} = \eta_1 \times \eta_2 \times \eta_3 \times \dots$ (each as a decimal) Then $\times 100$ to express as a percentage. The overall figure is always lower than the worst single step. Analogy — three middlemen each take a cut Say your pocket money passes through three people, and each keeps a slice. The first lets 90% through, the second 50%, the third 95%. You don't subtract their cuts — you keep

Source	Conversion chain	Typical overall efficiency
Hydro	Potential → Kinetic → Mech → Elec	~90%
Wind	Kinetic → Mech → Elec	~40%*
Gas (CCGT)	Chem → Thermal → Kinetic → Elec	~55%
Black coal	Chem → Thermal → Kinetic → Elec	~38%
Brown coal	Chem → Thermal → Kinetic → Elec	~28%
Solar PV	Light → Elec (1 step)	~20%

Build-your-own Sankey diagram

Interactive · Scene 03 Build-your-own Sankey diagram A Sankey diagram draws energy as a flow whose width is proportional to the amount. The useful stream stays on the main path; wasted heat branches off. Drag the three step efficiencies and watch the river narrow. Step 1 — burn (chem→thermal) 90% Step 2 — turbine (thermal→kinetic) 42% Step 3 — generator (kinetic→elec) 95% — Overall efficiency — Useful out (of 1000 MJ) — Wasted as heat Try dragging step 2 (the turbine) up to 100% — see how much the whole river fattens. That single step is the heat-engine

See the marks being earned

Worked examples See the marks being earned Three classic question types. Open each, cover the working, and try it first. TYPE A Multi-step efficiency from a chain › Yarrabin's gas plant converts chemical energy → thermal (88%) → kinetic (60%) → electrical (96%). Calculate the overall efficiency. (2 marks) $\eta = 0.88 \times 0.60 \times 0.96$ ← multiply the decimals $\eta = 0.507$ $\eta \approx 51\%$ Marking: 1 mark for multiplying (not adding) the three decimals; 1 mark for the correct answer with a % sign. Round sensibly and keep the unit. TYPE B Find the useful output from input + e

What the command word is really asking

◆ COMMAND WORDS

Decode the question What the command word is really asking In calculation questions, the verb tells you exactly how much working to show. Match your response to it. Calculate show working Reach a numerical answer with units . Write each line of working — marks live in the method, not just the final number. Determine show working Same as calculate — find the value, showing the steps that get you there. Show that → target You're given the answer. Your job is the working that lands on it. Don't skip steps; finish on the stated value. Compare both sides Stat

Six traps that cost real marks

⚠ EXAM TRAPS

Don't lose easy marks Six traps that cost real marks 1 Adding step efficiencies instead of multiplying 90% + 42% + 95% is meaningless (it's over 100%!). Fix: multiply the decimals — $0.90 \times 0.42 \times 0.95$. 2 Forgetting the $\times 100$, or doing it twice 0.36 is a fraction; 36% is a percentage. Decide once whether you're working in decimals or percent and stay there. Fix: work in decimals, convert to % only at the very end. 3 Flipping the ratio (input over output) Putting total input on top gives an efficiency above 100% — impossible. Fix: useful output always goes

The P·E·L move for "explain" parts

P·E·L WRITING

Writing trainer The P·E·L move for "explain" parts Calculation questions often end with a worded part — "explain why multi-step generation is less efficient." Examiners want a Point , Evidence , and a Link . Type an answer and watch the checklist light up. Explain why a coal-fired plant is less efficient overall than a hydro plant. P Point — state the claim E Evidence — forms & figures L Link — back to the question Point: more steps / heat loss Names a form (thermal / heat) Losses multiply / compound Heat-engine / thermal step Compares hydro (fewer st



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Five VCAA-style questions

★ PRACTICE & SELF-MARK

Practice exam · 20 marks Five VCAA-style questions Work each one on paper first. Then reveal the weak answer, the strong answer and the marking guide, and award yourself a score — it saves to your radial dock. 1 2 marks Calculate · single concept A wind turbine at Yarrabin receives 800 kJ of kinetic energy from the wind each second and outputs 320 kJ of electrical energy. Calculate its efficiency. ✗ Weak answer ✓ Strong answer ★ Marking guide ✗ Weak — 0–1 / 2 "800 – 320 = 480, so it's 480 kJ efficient." Confuses the wasted energy with efficiency, gives n

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

800 – 320 = 480, so it's 480 kJ efficient." Confuses the wasted energy with efficiency, gives no ratio, no percentage. This is a wasted-energy calculation, not an efficiency one.

✓ STRONG / MODEL ANSWER

$\eta = \text{useful output} \div \text{total input} \times 100 = 320 \div 800 \times 100 = 40\%$. Correct formula, substitution shown, units cancel, percentage stated.

✗ WEAK ANSWER

85 + 40 + 95 = 220, divided by 3 = 73% efficient. So $0.73 \times 2000 = 1460 \text{ MJ}$." Averaging the percentages is the classic multi-step error — efficiency can't be found by adding or averaging. Both the method and both numbers are wrong.

✓ STRONG / MODEL ANSWER

$\eta_{\text{overall}} = 0.85 \times 0.40 \times 0.95 = 0.323 = 32\% (\approx 32.3\%)$. Electricity out = $0.323 \times 2000 = \approx 646 \text{ MJ}$. Multiplies decimals, converts to %, then applies it to the input.

✗ WEAK ANSWER

The student is wrong because the plant is only 30% efficient, so 70% is lost. That's just how power stations work." Repeats the misconception ("lost"), names no forms, cites no law, and doesn't correct the "destroyed" claim. Only the efficiency figure earns anything.

✓ STRONG / MODEL ANSWER

The energy is not destroyed . The First Law of Thermodynamics states energy cannot be created or destroyed, only transformed. In the coal plant, chemical energy becomes thermal, then kinetic, then electrical. The "wasted" 70% is real energy that has been transferred to the surroundings as low-grade thermal energy (heat) — in the cooling water, flue gases and friction — at each conversion step. It is degraded to a less useful form, not gone. Corrects the error, names forms, cites the First Law, explains where the energy actually goes.

✗ WEAK ANSWER

Hydro is better because it's renewable and cleaner. Coal is bad for the environment and only 30% efficient. Hydro is 90%." Drifts into environmental impact (not asked), gives the figures but never uses the conversion chains or named forms to explain why the efficiencies differ. No comparison of mechanism.

✓ STRONG / MODEL ANSWER

Hydro chain: gravitational potential → kinetic (falling water) → mechanical (turbine) → electrical. There is no thermal step , so it avoids the big heat-engine loss and reaches ~90%. Coal chain: chemical → thermal → kinetic → electrical. The chemical-to-thermal-to-kinetic stage is a heat engine , which loses a large fraction as waste heat (limited by the Second Law), pulling the overall to ~30%. Because step efficiencies multiply , coal's extra thermal step compounds the losses, so hydro keeps roughly three times as much of its input as useful electricity. Both chains spelled out, named forms, identifies the thermal/heat-engine step as the difference, uses the multiply rule, comparative conc

✗ WEAK ANSWER

(a) Gas is about 90% because that's the biggest number. (b) Gas needs less energy. (c) Gas is better because it's more efficient." Picks a single step instead of multiplying, gives no input figures, and the evaluation has no reasoning or limitation. Efficiency alone ignores fuel type, emissions and renewability.

✓ STRONG / MODEL ANSWER

(a) $\eta_{\text{gas}} = 0.90 \times 0.62 \times 0.96 = 0.536 \approx 54\%$. (b) input = useful $\div \eta$. Gas: $40 \div 0.536 \approx 75$ MJ of gas. Solar: $40 \div 0.21 \approx 190$ MJ of sunlight. (c) By efficiency, gas wins — it needs far less input per unit of electricity. But efficiency alone is misleading: the solar input is free, renewable sunlight, whereas the gas input is a finite fossil fuel that releases CO_2 . A low-efficiency renewable can still be the more sustainable choice. Correct multi-step calc, rearranges to find input for both, balanced evaluation naming a real limitation of the efficiency metric.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.